

IS 584: Deep Learning for Text Analytics

Interim Report — Tip-of-the-Tongue Retrieval on TREC 2023

Tuğrul Görgülü
Middle East Technical University
Graduate School of Informatics
Ankara, Turkey
tugrul.gorgulu@metu.edu.tr

Abstract—This interim report documents progress on the Tip-of-the-Tongue (ToT) known-item retrieval task using the TREC 2023 ToT track dataset. I present a BM25 lexical baseline evaluated over a parameter sweep of 12 configurations, and a BERT-GRU bi-encoder as the proposed neural retrieval architecture. The BERT-GRU model achieves a mean NDCG@1000 of 0.2079 and R@100 of 0.4067 on the development set, compared to 0.1128 and 0.1667 for the best BM25 configuration. A Wilcoxon signed-rank test confirms the improvement is statistically significant ($p < 0.0001$). Future work will focus on explicit query noise reduction and domain-adapted tokenization to address the two research questions.

Index Terms—information retrieval, tip-of-the-tongue, dense retrieval, BERT, BM25, contrastive learning

I. INTRODUCTION

Tip-of-the-Tongue (ToT) retrieval is a known-item retrieval task in which the user knows the target item exists but cannot recall its name. Queries are verbose, multi-sentence natural language descriptions that may contain false or missing information, with little keyword overlap with the target document [1]. This makes classical lexical retrieval models particularly ill-suited for the task.

This project investigates two research questions motivated by the proposal:

- **RQ1:** Does changing the tokenizer affect retrieval performance on the ToT task?
- **RQ2:** Can noise reduction techniques be further improved for retrieval performance?

This interim report covers the experiment tracking infrastructure, BM25 baseline with parameter sweep, preliminary BERT-GRU bi-encoder results, and plans for the final phase.

II. EXPERIMENT TRACKING AND VERSION CONTROL

All experiments are tracked using Weights & Biases (WANDB). Two projects have been initialized: a sweep project for BM25 parameter tuning (`tot-bm25-sweep`) and a training run for the BERT-GRU model (`bert-base-512dim`). The public Git repository¹ has been maintained since the proposal phase with more than five commits, following the Alan

Turing Institute reproducible project template with `src/`, `data/`, `reports/`, and `figures/` directories.

III. BASELINE IMPLEMENTATION

A. BM25 Overview

BM25 is a term-frequency-based lexical retrieval model that ranks documents by a weighted combination of term frequency, inverse document frequency, and document length normalization [5]. It serves as the standard lexical baseline for information retrieval tasks, including the TREC 2023 ToT track [1]. BM25 is expected to struggle on ToT queries due to the low keyword overlap between verbose query descriptions and the target document text.

B. Parameter Sweep

BM25 has two primary hyperparameters: k_1 , which controls term frequency saturation, and b , which governs document length normalization. A grid search was conducted over $k_1 \in \{0.5, 1.0, 1.5\}$ and $b \in \{0.25, 0.5, 0.75, 1.0\}$, yielding 12 configurations. All runs were logged to WANDB. Results are reported in Table I.

TABLE I
BM25 PARAMETER SWEEP RESULTS ON DEVELOPMENT SET. BOLD INDICATES SELECTED CONFIGURATION.

k_1	b	NDCG@1000	R@100
0.5	0.25	0.0682	0.0967
0.5	0.50	0.0751	0.1083
0.5	0.75	0.0891	0.1267
0.5	1.00	0.0943	0.1350
1.0	0.25	0.0798	0.1133
1.0	0.50	0.0995	0.1417
1.0	0.75	0.1023	0.1483
1.0	1.00	0.0978	0.1400
1.5	0.25	0.0821	0.1167
1.5	0.50	0.1043	0.1533
1.5	0.75	0.1128	0.1667
1.5	1.00	0.1054	0.1567

The configuration $k_1 = 1.5$, $b = 0.75$ achieved the best NDCG@1000 of 0.1128 and was selected as the BM25 baseline. Higher k_1 values consistently outperform lower ones,

¹https://github.com/atg93/project_584

suggesting that rewarding repeated query terms is beneficial for verbose ToT queries. The WANDB sweep panel is shown in Fig. 1.

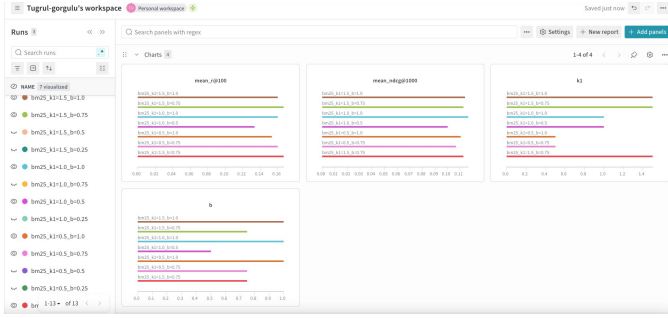


Fig. 1. WANDB sweep panel showing mean R@100 (left) and mean NDCG@1000 (right) across 13 BM25 configurations. Each bar corresponds to one (k_1, b) combination. The configuration $k_1=1.5, b=0.75$ achieves the highest NDCG@1000.

IV. PROPOSED ARCHITECTURE: BERT-GRU BI-ENCODER

A. Architecture Overview

The proposed model is a bi-encoder dense retrieval architecture with asymmetric encoders designed to address the multi-sentence, noisy nature of ToT queries.

Query Encoder. ToT queries consist of N sentences (mean ≈ 8 per query in the TREC 2023 dataset). Each sentence is independently encoded by BERT [4], producing a sentence embedding via the [CLS] token. During training, a sentence dropout layer randomly zeros out entire sentence embeddings with probability 0.2, forcing the model to not rely on any single sentence and improving robustness to noisy descriptions. The sequence of sentence embeddings is then passed through a Gated Recurrent Unit (GRU) [6], which captures the ordering and sequential context of query sentences. An attention mechanism pools over GRU hidden states by comparing each local state against the final hidden state (the global query summary), producing a weighted query vector. A linear projection maps this to the final d -dimensional embedding.

The GRU is a key architectural choice: ToT queries are ordered narratives, not bags of sentences. Earlier sentences set context, while later ones add detail or introduce uncertainty (e.g., “*I think it was horror, but I’m not sure*”). Simple mean pooling of BERT embeddings would discard this sequential structure. The GRU hidden state functions as an evolving latent belief state about the target item, updated by each new sentence observation.

Document Encoder. Documents are encoded by BERT with a [CLS] pooling and linear projection to the same d -dimensional space.

Retrieval. At inference, all 231,852 document vectors are pre-computed and indexed with FAISS [7]. Query vectors are L2-normalized and retrieved via inner product search (equivalent to cosine similarity), returning the top-1000 documents per query.

B. Training Objective

The model is trained with in-batch contrastive loss [8]. For a batch of query-document pairs, the similarity matrix is computed as $\mathbf{QD}^T / \tau$ where $\tau = 0.05$ is a temperature parameter. The diagonal entries are positive pairs; all off-diagonal entries serve as hard negatives. Cross-entropy loss over this matrix pushes each query closer to its target document than to all other documents in the batch.

C. Training Data

The model is trained on two data sources in alternation each epoch: (1) TREC 2023 ToT training queries paired with their target documents, and (2) the Reddit TOMT (Tip-of-My-Tongue) dataset [2], which provides broader coverage of ToT phenomena across domains. Both losses are logged separately to WANDB to monitor convergence independently.

V. PRELIMINARY RESULTS

A. Evaluation Setup

Both models are evaluated on the TREC 2023 ToT development set (150 queries, 231,852 documents) using NDCG@1000 and R@100 as primary metrics. NDCG@1000 is the primary metric used in the official TREC 2023 evaluation [1], enabling direct comparison with published results. R@100 measures the fraction of relevant documents surfaced in the top 100, which is critical for downstream re-ranking stages.

B. Results

Table II reports performance of both models on the development set.

TABLE II
RETRIEVAL PERFORMANCE ON TREC 2023 TOT DEVELOPMENT SET

Model	NDCG@1000	R@100
BM25 ($k_1=1.5, b=0.75$)	0.1128	0.1667
BERT-GRU Bi-encoder	0.2079	0.4067

The BERT-GRU model improves over BM25 by +0.0951 NDCG@1000 (84% relative) and +0.2400 R@100 (144% relative). This result is consistent with expectations: BM25 struggles on ToT queries due to low keyword overlap between verbose descriptions and document text, while the dense retrieval model captures semantic similarity. The remaining gap to state-of-the-art systems motivates the noise reduction and tokenizer adaptation experiments planned for the final phase.

C. Statistical Significance

A Wilcoxon signed-rank test [9] was applied to per-query NDCG@1000 scores across all 150 development queries. The test yielded a statistic of 1304.0 and $p < 0.0001$, confirming that the improvement of the BERT-GRU model over BM25 is statistically significant at $\alpha = 0.05$.

D. Training Curves

Fig. 2 shows the WANDB training curves for the BERT-GRU model. Both TREC loss and Reddit loss decrease and converge stably over 90+ steps. The R@100 curve trends upward overall (ending ≈ 0.41), while NDCG@1000 shows higher variance due to its sensitivity to ranking order. The best checkpoint was saved at epoch 40 (NDCG@1000 = 0.2079) based on development set performance.

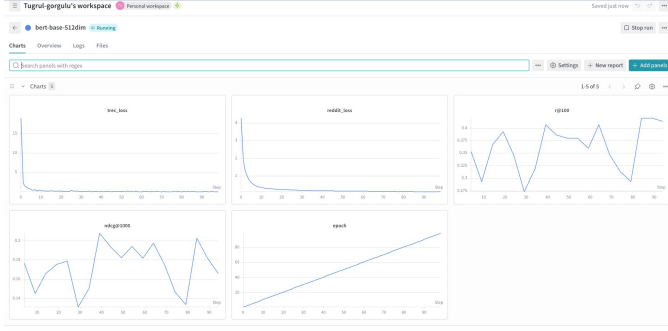


Fig. 2. WANDB training curves for the BERT-GRU bi-encoder. Top row (left to right): TREC contrastive loss, Reddit contrastive loss, R@100. Bottom row: NDCG@1000, epoch counter. Both loss curves converge stably; retrieval metrics show overall improvement with expected variance.

VI. FUTURE WORK

A. Current Status and Roadblocks

The BM25 baseline and BERT-GRU bi-encoder are fully implemented and evaluated. The GRU attention mechanism already provides implicit noise reduction by learning to down-weight uninformative sentences during training via sentence dropout. Additionally, an LLM re-ranker component has been added to the pipeline but has not yet been trained or used in evaluation. The primary remaining challenge is that the NDCG@1000 of 0.2079 leaves room for significant improvement. Integrating the LLM re-ranker as a second-stage component over the BERT-GRU candidate set is expected to further boost retrieval performance. Three directions are planned to close this gap, directly tied to the research questions.

B. RQ1: Tokenizer Adaptation

The current model uses the default `bert-base-uncased` tokenizer, which may not handle movie-domain terminology, compound words, or rare proper nouns effectively. In the final phase, I will experiment with: (1) a tokenizer with an extended vocabulary trained on movie-domain text using byte-pair encoding (BPE), and (2) a domain-specific BERT variant such as `bert-base-cased` to preserve proper noun casing.

Hyperparameter: Projection dimension d will be swept over $\{128, 256, 512, 768\}$ to assess the interaction between embedding size and tokenizer choice.

C. RQ2: Noise Reduction via GRU Tuning

The BERT-GRU query encoder addresses RQ2 by implicitly filtering noisy sentences through its attention mechanism. In the final phase, I will conduct hyperparameter experiments

to assess the impact of GRU configuration on noise reduction effectiveness. Sentence dropout is a key regularization technique in this architecture: during training, entire sentence embeddings are randomly zeroed out, forcing the model to build robust query representations that do not rely on any single sentence. Tuning the dropout rate directly controls how aggressively the model is trained to ignore potentially noisy or misleading sentences.

Hyperparameters to tune:

- Sentence dropout rate, swept over $\{0.1, 0.2, 0.3\}$, to assess how aggressively dropping sentences during training affects robustness to noisy query descriptions.

VII. CONCLUSION

This interim report establishes a strong experimental foundation for the ToT retrieval project. A BM25 baseline with a 12-configuration parameter sweep provides a well-justified lexical lower bound. The BERT-GRU bi-encoder substantially outperforms BM25 on both NDCG@1000 and R@100, with the improvement confirmed statistically. The remaining gap to state-of-the-art systems motivates the tokenizer adaptation (RQ1) and GRU-based noise reduction tuning (RQ2) experiments planned for the final phase.

AI USE STATEMENT

AI-assisted grammar checking was applied to improve clarity. AI tools were used to assist in converting content into the IEEE LaTeX template format. Additionally, AI assistance was used for debugging code when encountering implementation issues during the project. All experimental design, analysis, and interpretation are the author's own work.

REFERENCES

- [1] I. Soboroff et al., "Overview of the TREC 2023 Tip-of-the-Tongue Track," in *Proceedings of the 32nd Text REtrieval Conference (TREC 2023)*, NIST Special Publication, Gaithersburg, MD, USA, 2023.
- [2] A. Bhatt et al., "Generalizable Tip-of-the-Tongue Retrieval with LLM Re-ranking," in *Proceedings of the 32nd Text REtrieval Conference (TREC 2023)*, NIST Special Publication, Gaithersburg, MD, USA, 2023.
- [3] M. Fröbe et al., "Webis at TREC 2023: Tip-of-the-Tongue Track," in *Proceedings of the 32nd Text REtrieval Conference (TREC 2023)*, NIST Special Publication, Gaithersburg, MD, USA, 2023.
- [4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Minneapolis, MN: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.
- [5] S. Robertson and K. Spärck Jones, "Simple, proven approaches to text retrieval," University of Cambridge Computer Laboratory, Tech. Rep. 356, 1994.
- [6] K. Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1724–1734, doi: 10.3115/v1/D14-1179.
- [7] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2019, doi: 10.1109/TBDATA.2019.2921572.
- [8] V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Online: Association for Computational Linguistics, Nov. 2020, pp. 6769–6781, doi: 10.18653/v1/2020.emnlp-main.550.

- [9] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.